

Digital Egypt Pioneers Initiative (DEPI)

Graduation Project — Deliverable File 3

REQUIREMENTS GATHERING

Houra Platform

Stakeholder Analysis | User Stories & Use Cases | Functional & Non-Functional Requirements

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Date: 2026

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1. Introduction

This document presents the requirements gathering phase for the Time Banking Platform. It identifies the key stakeholders and their needs, translates those needs into user stories and use case scenarios, and formally defines the functional and non-functional requirements that will guide the System Analysis & Design phase.

The requirements below build directly on the problem, opportunity, and objectives already defined in the Project Proposal, and incorporate the detailed information-requirements analysis (Who / What / Where / How / Why) conducted by the team during the initial project study.

2. Stakeholder Analysis

The table below identifies the key stakeholders of the Time Banking Platform and summarizes what each group needs from the system in order for the project to succeed.

Stakeholder	Category	Needs & Expectations
Service Seekers	Primary User	Learn a new skill, language, or receive a service without financial cost; find trustworthy providers quickly.
Service Providers	Primary User	Offer their skills/time, be recognized for their expertise, earn time-credit fairly, and be protected from abuse of the system.
Platform Administrators / Moderators	Internal	Review and approve certificates/introduction videos, monitor exchanges, resolve disputes, and maintain trust and safety.
Academic Supervisor	Evaluator	Assess the project's completeness, technical quality, and adherence to the proposed scope and timeline.
Universities & Training Centers	Partner	Promote the platform to students, potentially co-host workshops, and support verification of academic credentials.
Non-Profit Organizations / Community Centers	Partner	Extend the platform's reach to underserved groups and support community adoption campaigns.

2.1 Verification Requirement (Cross-Stakeholder Need)

A recurring need across stakeholders is trust. To be accepted as a service provider, a user must be a serious, verifiable participant. Accordingly, the platform will require applicants to confirm their school/university affiliation, optionally upload a certificate relevant to the skill they wish to offer, and record a short (up to 5-minute) introductory video. This directly serves the needs of Service Seekers (confidence in provider quality), Administrators (a basis for approval), and the platform's overall trust model.

3. User Stories & Use Cases

3.1 User Stories

The user stories below translate stakeholder needs into concrete, testable statements that describe system behavior from each user's perspective.

ID	As a...	I want to...	So that...
US-01	Service Seeker	search for a specific skill or service by category	I can quickly find someone who can help me.
US-02	Service Provider	create a profile listing my skills, certificate, and a short introduction video	other users can trust my qualifications before requesting my service.
US-03	New User	start with a small number of trial time-credits	I can try the platform before offering or requesting larger exchanges.
US-04	Service Provider	have my time-credit balance automatically increased after completing a service	I am fairly compensated for my time.
US-05	Service Seeker	have time-credits automatically deducted after receiving a service	the exchange is tracked transparently without manual accounting.
US-06	User	rate and review the person I exchanged a service with	future users can trust the quality of that provider.
US-07	Administrator	review submitted certificates and introduction videos before approving a new provider	only serious, verified users can offer services on the platform.
US-08	User	record a video offline and upload it once I have an internet connection	I am not restricted by connectivity when preparing my profile.
US-09	User	receive notifications about new matching requests or messages	I never miss an opportunity to exchange a service.

3.2 Key Use Case Scenarios

The scenarios below describe, in narrative form, how the primary actors interact with the system for its most important flows. Formal UML use case diagrams will be developed in the System Analysis & Design file.

UC-01: Register & Get Verified as a Provider

Actor: New User (prospective Service Provider)

The user signs up, fills in their profile with the skills they can offer and the skills they want to learn, confirms their school/university affiliation, optionally uploads a supporting certificate, and records a short introduction video (which can be recorded offline and uploaded later). An administrator reviews the submission and approves or rejects it before the user can start offering services.

UC-02: Request a Service

Actor: Service Seeker

The user searches or browses available services by category, selects a provider, and sends a request. If accepted, the exchange is scheduled. Upon completion, the requester's time balance is deducted by the agreed number of hours and the provider's balance is credited accordingly.

UC-03: Offer a Service

Actor: Service Provider

The user lists a new skill/service on their profile. When a request is received, they accept or decline it. After delivering the service, they confirm completion, which triggers the time-credit transfer and prompts both users to rate each other.

UC-04: Rate & Review After an Exchange

Actor: Service Seeker / Service Provider

After an exchange is marked complete, both participants are prompted to rate the experience and leave an optional written review. Ratings are aggregated and displayed on each user's public profile.

4. Functional Requirements

Functional requirements are grouped below by system area. Each requirement is uniquely identified (FR-XX) for traceability into design, implementation, and testing.

Account Management & Verification

FR-01	The system shall allow users to register using an email address and basic profile information.
FR-02	The system shall allow users to submit a certificate and/or a short introduction video as part of provider verification.
FR-03	The system shall allow an administrator to approve or reject provider verification requests.

Time Wallet & Transactions

FR-04	The system shall issue new users a starting balance of time-credits upon successful verification.
FR-05	The system shall automatically deduct time-credits from a requester's balance upon confirmed service completion.
FR-06	The system shall automatically add time-credits to a provider's balance upon confirmed service completion.
FR-07	The system shall apply a demand-based weighting to time-credit costs for underrepresented service categories.

Service Discovery & Matching

FR-08	The system shall allow users to search and filter available services by category, keyword, and rating.
FR-09	The system shall allow users to send and respond to service requests.

Trust & Community

FR-10	The system shall allow users to rate and review each other after a completed exchange.
FR-11	The system shall display a user's average rating and verification status on their public profile.
FR-12	The system shall send notifications for new requests, messages, and status updates.
FR-13	The system shall allow in-app messaging between matched users.

Administration

FR-14	The system shall provide an admin dashboard to monitor transactions, review flagged users, and moderate content.
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5. Non-Functional Requirements

Non-functional requirements define the quality attributes the system must satisfy, independent of specific features.

Category	Requirement
Performance	The system shall respond to 95% of user requests within 2 seconds under normal load.
Security	User data and transaction records shall be encrypted in transit and at rest; authentication shall use secure password hashing.
Usability	A first-time user shall be able to complete registration and submit a verification request within 10 minutes without external help.
Reliability	The system shall maintain at least 99% uptime, excluding scheduled maintenance.
Scalability	The system architecture shall support growth from a single-campus pilot to multiple communities without a full redesign.
Availability	Core browsing and search features shall remain available even during partial service degradation.
Data Integrity	Time-credit transactions shall be atomic — a transfer must fully complete or fully fail, never leaving balances inconsistent.
Compatibility	The platform shall be accessible via modern web browsers and function correctly on both desktop and mobile screen sizes.

6. Conclusion

This Requirements Gathering file establishes a clear, traceable link between stakeholder needs, user stories, and formal functional and non-functional requirements. Together, they form the foundation for the next deliverable — System Analysis & Design — where these requirements will be translated into use case diagrams, an ER diagram, data flow diagrams, and the overall system architecture.